create a recombinant DNA or chimeric DNA.

4. Introduction of the recombinant vectors into host cells (Example: bacteria)
5. Multiplication and selection of clones containing the recombinant molecules
6. Expression of the gene to produce the desired product.

Cloning in the molecular biology sense (as opposed to cloning whole organisms) is the making of many copies of a segment of DNA, such as a gene. Cloning makes it possible to generate large amounts of pure DNA, such as genes, which can then be manipulated in various ways, including mapping, sequencing, mutating and transforming cells. An overview of cloning strategies in recombinant DNA technology is shown in Figure 12.23.

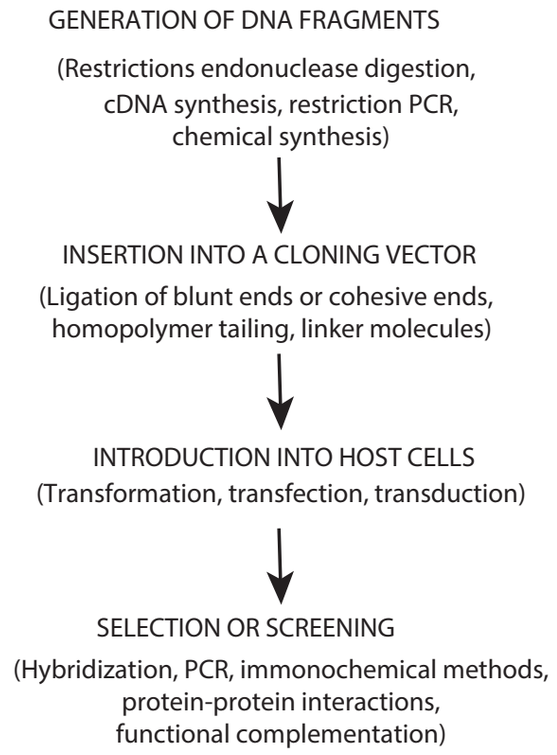


Figure 12.23: An overview of cloning strategies in recombinant DNA technology

12.9 Vectors, Types and Characteristics

Vectors are the DNA molecules, which carry a foreign DNA fragment to be cloned. They are cloning vehicles, examples of which are Plasmids, Bacteriophages, cosmids, phagemids and artificial chromosomes. The vector types differ in the molecular properties they have and in the maximum size of DNA that can be cloned into each.

Characteristics of an ideal vector.

1. Should be small in size
2. Should contain one or more restriction site
3. Should be self replicating
4. Should containing an origin of replication sequence (ori)
5. Should possess genetic markers (to detect the presence of vectors in recipient cells)

Plasmid Cloning Vectors

Bacterial plasmids are extra chromosomal elements that replicate autonomously in cells. Their DNA is circular and double stranded and carries sequences required for plasmid replication (ori sequence) and for the plasmid's other functions. (Note: A few bacteria contain linear plasmids. Example: *Streptomyces species, Borellia burgdorferi*). The size of plasmids varies from 1to 500 kb. Plasmids were the first cloning vectors. DNA fragments of about 570kb are efficiently cloned in plasmid cloning vectors. Plasmids are the easiest to work with. They are easy to isolate and purify, and they can be reintroduced into a bacteria by transformation. Naturally occurring plasmid vectors rarely possess all the characteristics of an ideal vector. Hence plasmid cloning vectors are

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